

Cor indep var aleatorias iff indep fn distribucion

Corolario 1. Sean $X_1, \dots, X_n$ variables aleatorias, son independientes

$$\iff \forall t_1, \dots, t_n \in \mathbb{R} : \mathbb{P}(\{X_1 \leq t_1\} \cap \dots \cap \{X_n \leq t_n\}) = \prod_{i=1}^n \mathbb{P}(X_i \leq t_i)$$

Demostración: La implicación $\Rightarrow$ es trivial por definición. Veamos la otra dirección:

$\Leftarrow$ Definimos $S_j = \{\{X_j \in (-\infty, t]\} : t \in \mathbb{R}\}$, entonces

$$\forall j \in \mathbb{N}_n : S_j \text{ es un } \pi\text{-sistema} \quad \wedge \quad S_1, \dots, S_n \text{ son independientes.}$$

Por la `prop-indep-pi-sistemas-imp-indep-sigma-algebras`/Proposición 1, $\sigma(S_1), \dots, \sigma(S_n)$ son independientes. Por otro lado, $\forall j \in \mathbb{N}_n : \sigma(S_j) = \sigma(X_j)$. Esto, por definición, implica que $X_1, \dots, X_n$ son independientes. ■

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